

COSET AND FIBER CONSTRUCTIONS FOR RAINBOW TREES IN FINITE ABELIAN GROUPS

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ABSTRACT. Let a rainbow tree over a finite abelian group G mean a tree on $|G|$ vertices whose vertices are bijectively labeled by G and whose edge sums are pairwise distinct. We give two exact composition principles. First, a rainbow tree over a subgroup $K \leq G$ admits a leaf extension over G for every independent choice of a permutation of K in each non-identity coset. Within the corresponding coset-respecting model, we classify all valid attachments and encode their degree increments by a circulant permanent. Second, if G/K has odd order, a rainbow tree over G/K and independently chosen rainbow trees over K can be assembled into a rainbow tree over G . This fiber construction works for nonsplit abelian extensions and for arbitrary missing colors in the fibers. Iteration produces multilevel and linear-maximum-degree families. These conditional constructions do not prove the unrestricted Graham–Sloane conjecture or its stronger bijective cyclic form.

1. INTRODUCTION

For a graph whose vertices are labeled in an abelian group, we call an edge *colored* by the sum of the labels at its endpoints. Graham and Sloane introduced harmonious labelings and conjectured that every tree is harmonious [1]. For a tree on $n \geq 2$ vertices, their classical convention uses $\mathbb{Z}_{n-1}$: every residue is used as a vertex label and exactly one label is repeated, while the $n - 1$ edge sums are all different. The stronger bijective cyclic formulation asks for a bijection to $\mathbb{Z}_n$ and $n - 1$ distinct edge sums. It implies the classical conjecture by deleting a leaf, labeling the remaining tree bijectively, and restoring the leaf with the label that realizes the unique missing edge color.

The October 2025 revision of the most recent primary paper located continues to formulate the unrestricted problems as conjectures and proves the stronger bijective statement for all sufficiently large trees of any fixed maximum degree [3]. Jamison and Kinnersley studied the group-valued version, including a direct-product construction, and gave a complete catalogue of failures for non-Boolean groups through order 20 [2].

This paper gives two elementary but flexible composition theorems. The first turns independent permutations in quotient cosets into leaf attachments and gives a converse classification within that model. The second replaces

the direct-product coordinates and common zero missing color in the closest earlier product construction by an arbitrary abelian extension with odd quotient and independently chosen fiber trees. The results occupy a degree regime not covered merely by fixing a maximum degree: one explicit family below has order n and maximum degree $n-4$. No claim is made for arbitrary input trees, arbitrary even quotients, or the unrestricted conjectures.

2. DEFINITIONS AND A BASIC OBSERVATION

Throughout, groups are finite and written additively.

Definition 2.1. Let G be a finite abelian group. A G -rainbow tree is a pair (T, f) in which T is a tree on $|G|$ vertices, $f : V(T) \rightarrow G$ is a bijection, and the colors

$$f(u) + f(v), \quad uv \in E(T),$$

are pairwise distinct. The unique group element not used as an edge color is the *free color* of (T, f) .

The free color exists because a tree on $|G|$ vertices has $|G| - 1$ edges. Translation of all vertex labels by $t \in G$ translates every edge color by $2t$ and therefore preserves the rainbow property, although it need not permit a prescribed free color when doubling is not surjective.

3. COSET-PERMUTATION LEAF EXTENSIONS

The first construction assigns a full block of new leaf labels to every nonidentity coset and lets a permutation determine their attachment points.

Theorem 3.1 (coset-permutation leaf extension). *Let $K \leq G$, and let (H, ϕ) be a K -rainbow tree. For each nonzero coset $C \in G/K$, choose a representative $c_C \in C$ and a permutation π_C of K . For every $k \in K$, introduce a vertex $z_{C,k}$ labeled $c_C + k$ and attach it as a leaf to the unique core vertex $v_{C,k}$ with*

$$\phi(v_{C,k}) = \pi_C(k) - k.$$

Together with the labels ϕ on H , this gives a G -rainbow tree.

Proof. The core labels exhaust K , and the new labels exhaust all other cosets, so the labeling is bijective. Adding leaves to H preserves the tree property. The color of the new edge incident with $z_{C,k}$ is

$$\phi(v_{C,k}) + (c_C + k) = c_C + \pi_C(k).$$

For fixed C , these colors exhaust C . Colors arising from different cosets are disjoint, while all core-edge colors lie in K and remain distinct. The core contributes $|K| - 1$ colors and the nonzero cosets contribute all their elements, for a total of $|G| - 1$ distinct colors. $\square$

The theorem includes concentrated and uniform attachment patterns, but its permutations give much finer control. The following converse records exactly how much control is available when the coset blocks and their leaf labels are fixed.

Proposition 3.2 (classification and degree enumerator). *Fix G, K, H, ϕ , the representatives c_C , and the new leaf labels $c_C + k$ as in Theorem 3.1. Allow the leaf indexed by k in C to be attached to the core vertex labeled $a_C(k)$, where $a_C : K \rightarrow K$ is arbitrary. The resulting labeling is rainbow if and only if*

$$k \mapsto k + a_C(k)$$

is a permutation of K for every nonzero coset C .

Put $r = |K|$ and $s = |G : K| - 1$, and introduce commuting variables x_a for $a \in K$. Then the attachment-degree increments are enumerated by

$$P_K(\mathbf{x})^s, \quad P_K(\mathbf{x}) = \text{per}(x_{b-a})_{a,b \in K} = \sum_{\pi \in \text{Sym}(K)} \prod_{k \in K} x_{\pi(k)-k}.$$

In particular, there are $(r!)^s$ valid ordered attachment choices. If a monomial has exponent vector $(d_a)_{a \in K}$, then necessarily

$$\sum_{a \in K} d_a = sr, \quad \sum_{a \in K} ad_a = 0 \quad \text{in } K.$$

Proof. In the coset C , the leaf-edge colors have K -coordinates $k + a_C(k)$. They are distinct exactly when this map is a permutation. The choices for different cosets are independent, giving $(r!)^s$ choices. Expanding the permanent selects one permutation and records each of its displacements; multiplying s copies records the ordered choices over all nonzero cosets. Every permutation has total displacement $\sum_k (\pi(k) - k) = 0$, which proves the two necessary identities. $\square$

Corollary 3.3 (selected cyclic profiles). *Let $K = \mathbb{Z}_r$. A translation $\pi(k) = k + a$ attaches all r leaves of its coset at the vertex labeled a . If r is odd, $\pi(k) = 2k$ attaches one leaf at each core vertex. If r is even, no permutation in this fixed model attaches exactly one leaf at each core vertex. More generally, if $\pi(k) = uk + a$ with u a unit modulo r and $d = \gcd(u - 1, r)$, then precisely r/d core vertices receive leaves and each receives d .*

Proof. The attachment label is the displacement $\pi(k) - k$. The affine displacement $(u - 1)k + a$ has image size r/d and every fiber has size d . For a uniform profile the displacements would be a permutation; when r is even, their sum would be $r/2$ in $\mathbb{Z}_r$, whereas the displacement sum of every permutation is zero. $\square$

For a general group K , a uniform coset block is the same as a complete mapping: both $k \mapsto \pi(k)$ and $k \mapsto \pi(k) - k$ must be permutations. By the Hall–Paige theorem, such a mapping exists precisely when the Sylow

2-subgroups of K are trivial or noncyclic [4]. This recovers the odd/even dichotomy above when K is cyclic.

Theorem 3.1 can be iterated along any subgroup chain. In the cyclic case, if $n_0 \mid n_1 \mid \cdots \mid n_t$ and the initial tree is $\mathbb{Z}_{n_0}$ -rainbow, repeated application gives a $\mathbb{Z}_{n_i}$ -rainbow tree at each stage. Later extensions may attach leaves to vertices introduced at earlier stages, producing multilevel branching.

As a concrete high-degree illustration, consider the subdivided claw with edges

$$01, 02, 03, 14, 25, 36.$$

The vertex labels $(0, 6, 1, 2, 4, 3, 5)$ in order $0, \dots, 6$ give the edge colors $(6, 1, 2, 3, 4, 0)$ in $\mathbb{Z}_7$. For every $q \geq 1$, a concentrated application of Theorem 3.1 attaches $7(q-1)$ leaves at vertex 0. The resulting noncaterpillar tree has order $n = 7q$ and maximum degree $n - 4$. This family illustrates the available degree range; it is not asserted as a separate novelty claim.

4. AN ODD-QUOTIENT FIBER CONSTRUCTION

The next theorem replaces each quotient vertex by an independently chosen rainbow fiber. The joining edges fill all but one of the free colors left by those fibers.

Theorem 4.1 (odd-quotient fiber construction). *Let G be a finite abelian group, let $K \leq G$, and suppose that $Q = G/K$ has odd order. Let (R, ρ) be a Q -rainbow tree, with the vertices of R identified with their labels $x \in Q$. For every $x \in Q$, let (S_x, σ_x) be an independently chosen K -rainbow tree, and let $\mu_x \in K$ be its free color. Then the fiber trees can be joined by $|Q| - 1$ edges to form a G -rainbow tree. This holds without assuming that G splits as a direct product and without requiring the free colors μ_x to agree.*

Proof. Choose a representative $c_x \in G$ of every $x \in Q$. In the copy of S_x , label v by $c_x + \sigma_x(v)$. These labels exhaust the coset $c_x + K$, so all fiber labels together exhaust G . The internal colors in this fiber are

$$(2c_x + K) \setminus \{m_x\}, \quad m_x := 2c_x + \mu_x.$$

Because $|Q|$ is odd, doubling is a bijection of Q . Hence the internal color cosets $2x$, as x ranges over Q , are disjoint and exhaust Q , with exactly one hole m_x in each.

For an edge $xy \in E(R)$, let $z \in Q$ be the unique element satisfying $2z = x + y$. Join a vertex $u \in S_x$ to a vertex $v \in S_y$ so that the new edge has color m_z . Such endpoints exist: the required equation is

$$\sigma_x(u) + \sigma_y(v) = 2c_z + \mu_z - c_x - c_y,$$

whose right-hand side belongs to K . Choose either summand arbitrarily; bijectivity of the other fiber labeling supplies the other endpoint.

The quotient edge colors $x+y$ are distinct, and halving is bijective on Q , so the indices z are distinct. The joining edges therefore fill distinct holes, one for every edge of R , leaving exactly one global free color. Finally, contracting

every fiber turns the constructed graph into R . Thus the $|Q| - 1$ joining edges connect the fiber trees without a cycle, and the resulting graph is a tree. Its vertex labels are bijective and all $|G| - 1$ edge colors are distinct. $\square$

The boundary cases are exact: a trivial quotient returns the sole fiber, and a trivial subgroup returns the quotient tree. Oddness is used precisely to make doubling invertible on Q .

In the split case $G = B \times C$ with $|B|$ odd, the construction at the end of Section 3 of Jamison and Kinnersley [2] uses a B -rainbow outer tree and C -rainbow fibers with common free color zero, joined at their zero-labeled vertices. Theorem 4.1 recovers that conclusion. Its endpoint equation is what permits both a nonsplit extension and arbitrary, independently varying free colors.

The theorem also iterates: whenever a chain of subgroups has successive odd-order quotients and rainbow input trees are available at each level, the fiber operation composes them into a rainbow tree over the top group. The input fibers need not be mutually isomorphic.

5. SCOPE AND REPRODUCIBILITY

The two constructions are conditional: they require already rainbow core, quotient, or fiber trees. They do not supply such a labeling for an arbitrary tree, and Theorem 4.1 makes no assertion for even-order quotients. In particular, neither theorem resolves the unrestricted classical or bijective cyclic harmonious-tree conjecture.

Solver-assisted experiments were used only as regression tests for the formulas, not as premises of any proof. The accompanying project repository contains a deterministic certificate verifier, exhaustive certificates for the project's cyclic computations through order 15, and direct tests of the two constructions. The order-15 cyclic computation is not claimed as new: Jamison and Kinnersley's catalogue through order 20 includes the cyclic cases [2]. Exact commands and counts are recorded with the release materials.

COMPUTATIONAL VERIFICATION

SAT search and deterministic scripts were used to test finite instances and to stress-test the explicit formulas. The theorem proofs are independent of these calculations.

REFERENCES

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